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PMHT: prompt-based spectral–spatiotemporal feature fusion with 2DMamba for hyperspectral object tracking

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ABSTRACT

Hyperspectral images provide rich spectral information that is advantageous for object tracking in challenging scenarios such as complex backgrounds, deformation, and occlusion. However, the limited availability of annotated hyperspectral data often forces existing methods to rely on RGB pre-trained models, resulting in suboptimal spectral representation and localization performance. To address this issue, this letter proposes a prompt-based multi-modal hyperspectral tracking framework (PMHT) with a 2D selective scanning mamba (2DMamba) backbone. A dual-stream prompting mechanism is designed to jointly exploit spectral and spatiotemporal information. Specifically, a neighbourhood group normalized matched filter is adopted to select three representative bands for spectral prompting, while inter-frame differences are utilized to construct a spatiotemporal prompt capturing temporal variations. These prompts are integrated with backbone features via a lightweight prompt feature fusion module to enhance the representation capability. In addition, a direction-aware localization loss with bounding-box shape constraints is introduced to improve spatial accuracy. Experiments on the HOTC2024 dataset demonstrate that the proposed method achieves state-of-the-art performance with high efficiency. The source code is available at <https://github.com/xuliny/PMHT>.

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Hyperspectral Videos;
Spectral–spatiotemporal;
Mamba; Prompt

1. Introduction

Visual object tracking, as a fundamental task in computer vision, has significant application value in fields such as security surveillance, autonomous driving, and remote sensing. In recent years, advances in hyperspectral imaging technology have enabled hyperspectral videos (HSVs) to emerge as a new medium for hyperspectral object tracking (HOT) (Gan et al. 2024; Xiong, Zhou, and Qian 2020). The unique narrow-band spectral information in HSVs can reveal the material properties of targets, offering notable advantages over RGB-based trackers in challenging scenarios (Wang et al. 2025; Xiang et al. 2026) involving complex backgrounds, target deformations, and occlusions.

Currently, deep learning-based frameworks dominate the field of hyperspectral object tracking. However, due to the limited availability of annotated hyperspectral training data (Zhao et al. 2022), most existing deep learning-based hyperspectral trackers.

Rely on pre-trained RGB trackers trained on large-scale RGB tracking datasets (e.g., LaSOT, GOT-10k) (Liu et al. 2022). Some of these trackers convert hyperspectral images into a single three-channel false-colour image through band selection (Tang et al. 2022), while others generate multiple three-channel false-colour images by recombining the bands (Gao et al. 2023; W. Li et al. 2023), in order to fit the adapted tracking networks. Such approaches either fail to fully utilize the spectral detail or introduce redundancy and increased computational complexity.

In recent years, prompt learning models (Y. Chen, Tang, et al. 2024; Zhu et al. 2023) have gained increasing attention in deep learning, emerging as a promising approach to improve model generalization and adaptability. To address the limitations of insufficient hyperspectral training data while improving the utilization of spatiotemporal information, we incorporate the concept of prompt learning and propose PMHT (prompt-based multi-modal hyperspectral tracking). Our method utilizes a pre-trained RGB tracker as the base model for hyperspectral object tracking. PMHT freezes the entire base model and only learns modality-specific visual prompts, avoiding the fine-tuning of the RGB tracker, thus maximizing the inheritance of its learned parameters. In recent years, the Mamba architecture (Gu and Dao 2023) based on state space models has received significant attention due to its linear computational complexity and strong feature modelling capabilities when processing long-sequence data. Inspired by this, we introduce the 2DMamba structure (J. Zhang et al. 2025) as the core component of the auxiliary branch, aiming to leverage its efficient parallel processing capability and strong adaptability to multi-dimensional data. This design enhances the expressive quality of the prompt features, thereby effectively improving the accuracy and robustness of the overall tracking model.

Since the pre-trained RGB tracker takes three-channel images as input, we first introduce the CIE color matching (CMFs) to transform the hyperspectral image into a three-channel false-colour image, which serves as the base input to the backbone network. Meanwhile, the neighbourhood group normalized matched filter (NGNMF) (Fu et al. 2022) is applied to select the three most representative spectral bands from the entire hyperspectral cube, which are then combined into a false-colour image and fed into the tracker as spectral prompt information. In addition, to enhance the model's ability to capture spatio-temporal information, processed adjacent frames are introduced as spatiotemporal prompts. The prompt feature fusion module (PFFM) subsequently integrates the base features with the spectral and spatio-temporal prompts, enriching the feature representation and improving tracking performance. Finally, to further improve localization accuracy and generalization, we incorporate a direction-aware mechanism and a penalty term for a shape of the bounding box in the loss function (Gevorgyan 2022), strengthening the model's ability to learn directional offsets and enforce geometric consistency constraints, as shown in Figure 1.

The contributions of this work are summarized as follows:

- A novel hyperspectral object tracking framework using prompt learning is proposed, featuring spectral and spatio-temporal prompt branches to improve performance in challenging scenarios like complex backgrounds and fast target changes.
- An efficient PFFM using the 2DMamba architecture is designed to extract features and adapt structurally in modelling multi-dimensional data. It leverages spatial details and spectral information from HSVs to produce discriminative features.

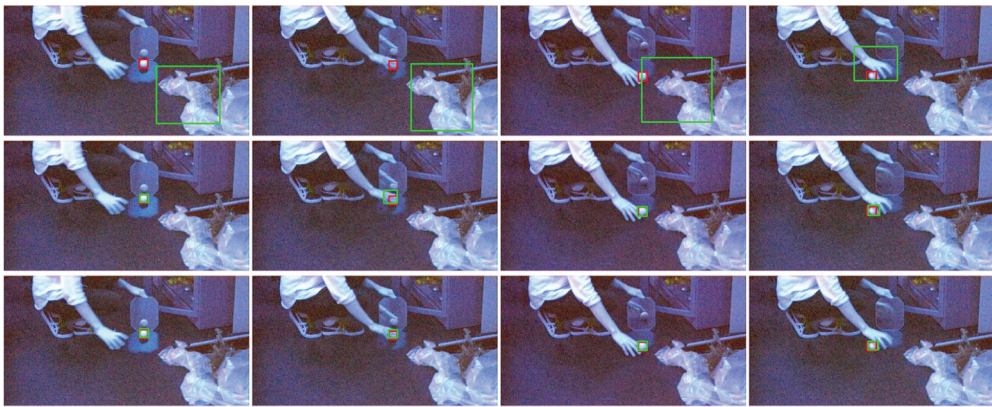


Figure 1. The model significantly improves in fast motion, object interference, and occlusion scenarios. (a) The first row shows the baseline model without PFFM, which suffers from severe tracking drift and loses the target when the object moves rapidly. (b) The second row demonstrates that adding the Prompt Generation Module significantly improves tracking stability, with noticeable improvements observed across all four images. (c) The third row presents the full model incorporating the 2DMamba module, which achieves more stable tracking compared to the version with only the prompt generation module, as shown in the second column of the third row. (Red: groundtruth; Green: prediction).

- The NGNMF technique selects the three most representative spectral bands from the hyperspectral cube to enhance spectral information utilization efficiency. In addition, the loss function is enhanced for better localization and generalization by integrating a direction-aware mechanism and a bounding box shape penalty.

The main subjects covered in this article are outlined below. [Section 1](#) contains the introduction. [Section 2](#) reviews the related work. [Section 3](#) introduces the proposed tracking algorithm. [Section 4.1](#) presents the experimental findings and their analysis. Finally, [Section 5](#) wraps up the article.

2. Related work

2.1. Hyperspectral object tracking

Hyperspectral images provide rich spectral information that enables effective discrimination of objects with similar visible reflectance and exhibit strong robustness to illumination variations and shadow interference, making them particularly suitable for target recognition in complex scenes. Early hyperspectral video tracking methods mainly relied on hand-crafted features (Hou et al. 2022). Xiong, Zhou, and Qian (2020) proposed a material-based tracking framework using spectral-spatial feature extractors, while Qian et al. (2018) combined convolutional networks with kernel correlation filters by employing 3D spectral cubes as convolution kernels. However, hand-crafted features are prone to tracking drift due to limited target representation capability.

To address this issue, deep learning-based hyperspectral trackers (J. Chen et al. 2025; Qian et al. 2025) have been widely explored. SiamHYPER (Liu et al. 2022) leverages pre-

trained Siamese networks to achieve robust tracking with limited hyperspectral samples, while SEE-Net (Z. Li et al. 2023) models spectral correlations through a self-expression mechanism. More recently, transformer-based approaches have demonstrated strong potential. Trans-DAT (Wu et al. 2024) introduces banded spatial attention to better exploit spectral information. Despite these advances, hyperspectral target tracking still suffers from band redundancy and insufficient feature utilization, and most existing methods overlook the importance of contextual spatiotemporal information.

2.2. Prompt-based learning in vision

Traditionally, large-scale pre-trained models are adapted to downstream tasks through full-parameter fine-tuning, which is computationally expensive and prone to overfitting when task-specific data are limited. Inspired by the success of prompt learning in natural language processing, recent studies have introduced prompt-based paradigms into vision tasks, where task-specific prompts guide model adaptation and improve contextual understanding. Jia et al. (2022) first proposed Visual Prompt Tuning (VPT) as an alternative to full fine-tuning for Vision Transformers, achieving superior tracking performance with significantly fewer trainable parameters. This paradigm has since been validated in various vision tasks, including image classification, segmentation, and cross-modal learning (Z. Zhang et al. 2025).

In visual tracking, ProTrack (Yang et al. 2022) pioneered the use of prompt learning by unifying multimodal inputs into a single modality, while ViPT (Zhu et al. 2023) further improved tracking robustness by integrating modality-aware prompts into large pre-trained models. However, existing prompt-based trackers do not fully exploit hyperspectral information and often fail to learn discriminative spectral representations. To address this limitation, our tracker incorporates CMFs-generated band selection cues as input to guide the NGNMF. This collaborative design combines NGNMF's spatial sensitivity with density-based band selection's statistical robustness, allowing accurate target localization while reducing local optimization information loss.

3. The proposed algorithm

3.1. Overview

3.1.1. Input

The input to the PMHT network (in Figure 2) consists of three components: two consecutive frames of hyperspectral images, denoted as $\mathbf{H}_t \in \mathbb{R}^{N \times H \times W}$ and $\mathbf{H}_{t-1} \in \mathbb{R}^{N \times H \times W}$, and a template hyperspectral image $\mathbf{H}_0$. Here, N represents the number of spectral bands, while H and W denote the height and width of the images, respectively. Specifically, for the current frame $\mathbf{H}_t$, we apply the CMFs to generate a dimensionality-reduced false-colour image $\mathbf{F}_t \in \mathbb{R}^{3 \times H \times W}$. Subsequently, the NGNMF (Fu et al. 2022) is used to select the three most representative spectral bands from $\mathbf{H}_t$, which are then used to construct another false-colour image $\mathbf{P}_t \in \mathbb{R}^{3 \times H \times W}$. Similarly, the previous frame $t-1$ is processed using CMFs to obtain its corresponding false-colour representation $\mathbf{F}_{t-1} \in \mathbb{R}^{3 \times H \times W}$. In this way, the three images are fed into the patch embedding layer, where each input stream is independently projected and flattened into the latent space to produce the

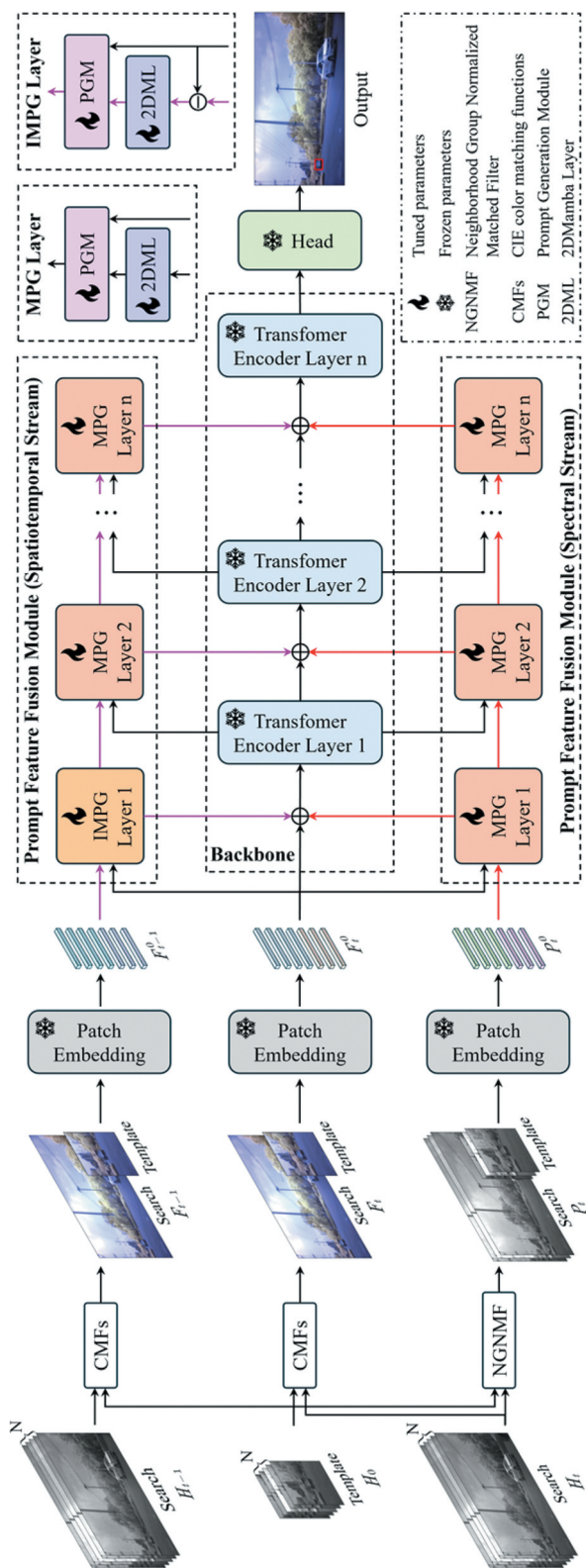


Figure 2. The flowchart of the proposed algorithm. (MPG: prompt generation with 2DMamba module. IMPG: init prompt generation with 2DMamba module).

corresponding token sequences. The initial token extracted from $\mathbf{F}_t$, $\mathbf{P}_t$, and $\mathbf{F}_{t-1}$ are denoted as $\mathbf{F}_t^0$, $\mathbf{P}_t^0$, and $\mathbf{F}_{t-1}^0$, respectively.

3.1.2. Prompt feature fusion module

The proposed PFFM is organized into two distinct streams: the spatio-temporal stream and the spectral stream. The initial tokens $\mathbf{F}_t^0$ and $\mathbf{F}_{t-1}^0$ are fed into the spatio-temporal stream, while $\mathbf{F}_t^0$ and $\mathbf{P}_t^0$ are input into the spectral stream. The spatio-temporal stream comprises one init prompt generation with 2DMamba (IMPG) module followed by multiple prompt generation with 2DMamba (MPG) modules, whereas the Spectral Stream consists solely of MPG modules. Both the IMPG and MPG modules integrate a 2DMamba layer (2DML) and a prompt generation module (PGM).

3.1.3. Base model

The model adopts ViT (Dosovitskiy et al. 2020) and OTrack (Ye et al. 2022) as base architectures. In each layer, the spatiotemporal prompt information and spectral prompt information from the previous layer are fused with the base features and then fed into the encoder of the current layer. The output of the final encoder layer is first passed through a linear layer to reduce the channel dimension, and then input to the prediction head for classification and object bounding box regression. The PMHT is initialized with the parameters of the base model, OTrack. The loss l_{track} , apart from the target box loss, generally follows the form of the base framework.

$$l_{\text{track}} = L_{\text{cls}} + \lambda_{L_1} L_1 + \lambda_{\text{iou}} L_{\text{SIOU}} \quad (1)$$

where L_{cls} is the weighted focal loss, L_1 and the SCYLLA-IoU (Gevorgyan 2022) loss L_{SIOU} jointly optimize the bounding box regression, and λ_{L_1} and λ_{iou} are balancing coefficients.

3.2. Prompt feature fusion module

In Figure 2, the proposed PFFM architecture is divided into a spatio-temporal stream and a spectral stream. These differ primarily in their module composition – the spatiotemporal stream begins with an IMPG module, while the Spectral Stream is entirely composed of MPG modules. The key difference in the IMPG module lies in the early fusion of base information from the current frame and the previous frame prior to the 2DML module, which enables the extraction of spatio-temporally correlated prompt features.

3.2.1. 2DML

Mamba achieves efficient modelling of long sequences through the use of a State Space Model and a selective scanning mechanism. However, the one-dimensional scanning strategy of traditional Mamba leads to a loss of geometric structure information when processing two-dimensional spatial data. Thus, this paper introduces the 2DMamba module, which employs a two-dimensional selective scanning mechanism to independently enhance spatio-temporal and spectral prompts, optimizing cross-modal feature fusion in target tracking.

In Figure 3, the key component of 2DMamba is the 2D selective scan operation. Given an input feature map $\mathbf{M} \in \mathbb{R}^{H \times W \times C}$, 2DMamba models spatial dependencies through

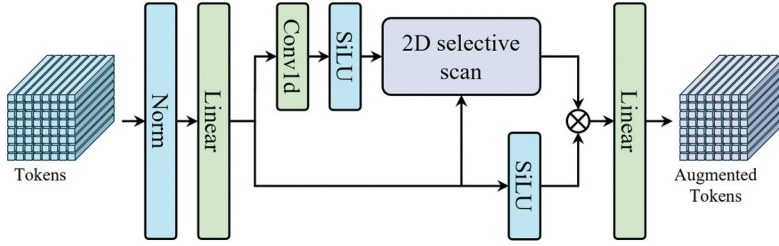


Figure 3. Detailed design of the proposed 2DML.

horizontal and vertical scanning. For horizontal scanning, the local context is recursively aggregated along the row direction for each column.

$$\mathbf{H}_{i,j}^{\text{hor}} = \bar{\mathbf{A}}_{i,j} \mathbf{H}_{i,j-1}^{\text{hor}} + \bar{\mathbf{B}}_{i,j} \mathbf{M}_{i,j} \quad (2)$$

Here, $\mathbf{H}_{i,j-1}^{\text{hor}}$ denotes the hidden state of the previous column ($i, j-1$) in the same row, with $\bar{\mathbf{A}}_{i,j}$ controlling the decay rate of historical information and $\bar{\mathbf{B}}_{i,j}$ regulating the weight of the current input $\mathbf{M}_{i,j}$. The superscript hor indicates that this hidden state has undergone only horizontal scanning.

For vertical scanning, global dependencies are fused along the column direction for each row based on the results from horizontal scanning. The hidden state $\mathbf{H}_{i,j}$ is updated as

$$\mathbf{H}_{i,j} = \bar{\mathbf{A}}_{i,j} \mathbf{H}_{i-1,j} + \mathbf{H}_{i,j}^{\text{hor}} \quad (3)$$

where $\mathbf{H}_{i-1,j}$ is the hidden state of the previous row ($i-1, j$) in the same column. Parameters $\bar{\mathbf{A}}_{i,j}$ and $\bar{\mathbf{B}}_{i,j}$ are dynamically generated based on the input feature $\mathbf{M}_{i,j}$, ensuring that the model adaptively screens target-sensitive regions.

Using a dual-scanning mechanism, 2DMamba preserves the two-dimensional spatial topology, avoiding the disruption of spatial neighbourhood relationships caused by traditional one-dimensional scanning. This precise modelling of targets across spatiotemporal and spectral dimensions is further enhanced by the dynamic parameter $(\bar{\mathbf{A}}_{i,j}, \bar{\mathbf{B}}_{i,j})$, which suppress background noise while focusing on target features.

3.2.2. PGM

To generate effective visual prompts for tracking, we employ PGM, as illustrated in Figure 4. It consists of three core components: the Channel Attention Module, the Token Attention Module, and the cross-fusion module (CFM).

The channel attention module enhances target-related features by dynamically allocating channel-wise weights. Given an input feature $\mathbf{T} \in \mathbb{R}^{D \times H \times W \times K}$, the global average pooling and the max pooling are applied along the spatial dimensions to obtain $\mathbf{Z}_{\text{avg}}^c$ and $\mathbf{Z}_{\text{max}}^c$, respectively. A lightweight convolutional layer with a non-linear transformation is then used to generate the channel weight map, and the enhanced feature $\mathbf{T}_c$ is obtained by

$$\mathbf{T}_c = \mathbf{T} \cdot \sigma \left(\mathcal{F}_c(\mathbf{Z}_{\text{avg}}^c) + \mathcal{F}_c(\mathbf{Z}_{\text{max}}^c) \right) \quad (4)$$

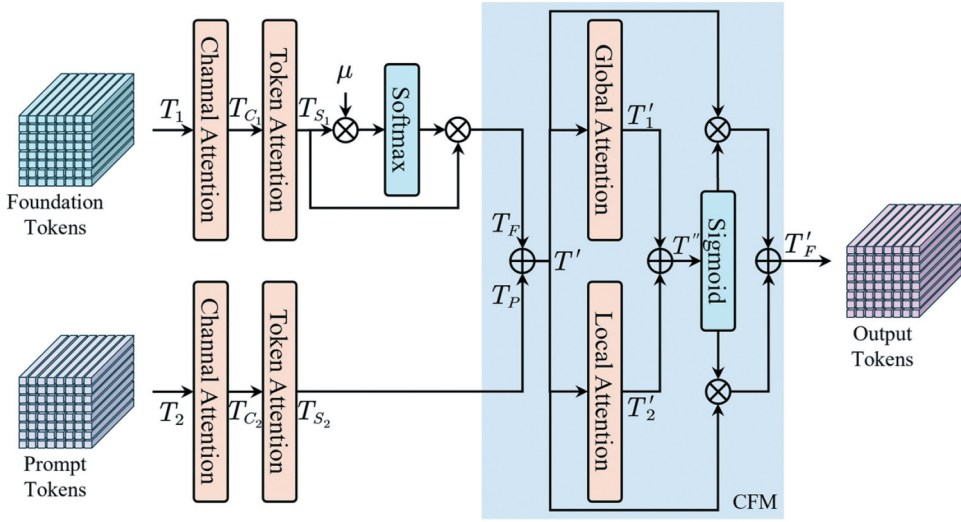


Figure 4. Detailed design of the proposed PGM. (CFM: Cross-fusion module).

where $F_c(\cdot)$ denotes the operation of channel squeezing and excitation, and σ represents the sigmoid function.

The token attention module focuses on salient regions of the target. For an input feature $T_c \in R^{D \times H_k \times W_k}$, average pooling and max pooling are performed along the channel dimension. The resulting feature maps are concatenated and fed into a convolutional layer to generate the spatial weight map, and the enhanced feature T_s is,

$$T_s = T_c \cdot \sigma\left(\mathcal{F}_s(\mathcal{C}(\mathbf{Z}_{\text{avg}}^s, \mathbf{Z}_{\text{max}}^s))\right) \quad (5)$$

where $F_s(\cdot)$ and $C_s(\cdot)$ represent the spatial convolution operation and the concat operation, respectively.

The CFM enables deep fusion of multi-modal features through a dual-branch interaction. Given two input features T_F and T_P , their sum is passed through both a pooling branch and a convolutional branch to extract complementary features. These features are then combined to produce a dynamic weight map T' , and the final result is T'_o obtained by weighted summation.

$$T'_o = T_F \cdot \sigma(T') + T_P \cdot (1 - \sigma(T')) \quad (6)$$

In the foundation token branch, we employ a learnable parameter to dynamically calibrate the feature response distribution, allowing the model to adaptively enhance high-confidence structural regions of targets while suppressing interference signals from background areas, thus achieving refined feature space decoupling.

Subsequently, the spatiotemporal and spectral prompts are fused with the backbone network base tokens, which drives the encoder to generate robust target representations.

4. Experimental results and analysis

4.1. Experimental setup

The data set used in this study was provided by the 2024 hyperspectral object tracking challenge (HOTC2024) (Xiong et al. 2024). It consists of three subsets: NIR, RedNIR, and VIS, which contain 25, 15, and 16 spectral bands, respectively. The HOTC2024 data set includes false colour videos corresponding to HSVs, generated using CIE colour matching functions (Xiong, Zhou, and Qian 2020). NIR contains 70 training videos and 30 testing videos, RedNIR contains 36 training videos and 20 testing videos, and VIS contains 111 training videos and 67 testing videos. All experiments were carried out using the PyTorch framework on an NVIDIA GeForce RTX 4090 GPU. The batch size is set to 24, and the AdamW optimizer with a weight decay of 10^{-4} is used. The learning rate is fixed at 4×10^{-5} . The model is trained for 30 epochs. The fixed parameters are initialized from a pre-trained base model, OTrack, while the trainable feature learning parameters are initialized using the Xavier uniform initialization scheme. To evaluate the tracking performance, we utilize the Distance Precision at 20 pixels (DP@20P) and the area under the curve (AUC) as the primary metrics.

4.2. Qualitative comparison

To further demonstrate the superior performance of our tracker, Figure 5 presents visual comparison results on the HOTC2024 dataset among PMHT and other SOTA HOT methods, including SPIRIT (Y. Chen et al. 2023), PHTrack (Y. Chen, Tang, et al. 2024), AQATrack (Xie et al. 2024), TransDAT (Wu et al. 2024), SENSE (Y. Chen, Yuan, et al. 2024), and SSTtrack (Y. Chen et al. 2025). The scenes in Figure 5 are selected from three subsets of HOTC2024: NIR, RedNIR, and VIS, specifically from the following videos: NIR-car43, RedNIR-ball&mirror10, VIS-S warmup, and VIS-dice3. As shown in Figure 5, PMHT outperforms the other trackers under challenging scenarios such as

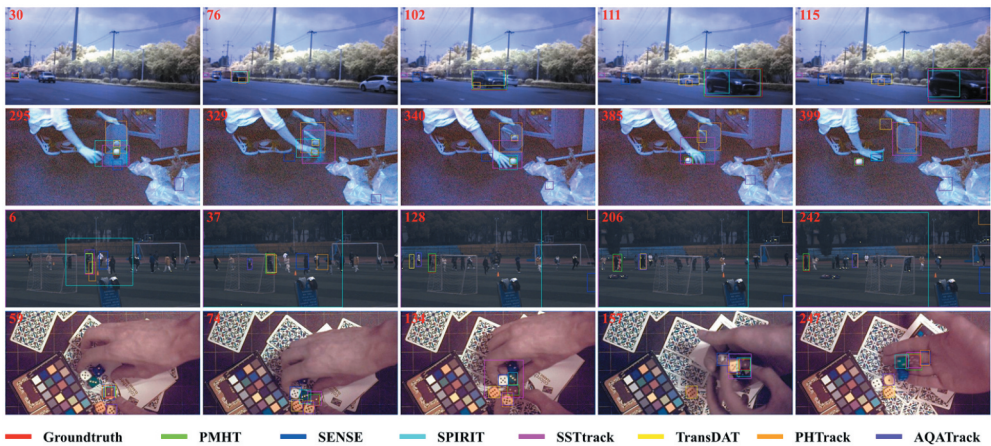


Figure 5. Visualization comparisons of PMHT with other SOTA HOT methods on four video sequences (i.e., NIR-car43, RedNIR-ball&mirror10, VIS-S warmup, and VISdice3) from the HOTC2024 dataset are provided.

rapid target scale variation (e.g., NIR-car43), fast target movement (e.g., RedNIRball&mirror10), target occlusion (e.g., VIS-S warmup), and presence of distractors (e.g., VIS-dice3). In summary, the qualitative results validate the effectiveness of the proposed method in PMHT and demonstrate its robustness and accuracy in HOT.

4.3. Quantitative comparison

We compare the proposed PMHT method with six state-of-the-art hyperspectral trackers, including SPIRIT, PHTrack, AQATrack, TransDAT, SENSE, and SSTrack. As shown in Figure 6 and Table 1, PMHT achieves the best performance in the entire HOTC2024 data set, achieving a DP@20P score of 0.707 and an AUC score of 0.543. This demonstrates the effectiveness of PMHT in HOT.

4.4. The ablation study

To evaluate the effectiveness of the proposed algorithm, ablation experiments were conducted by removing specific components and testing on the same hyperspectral dataset, with performance assessed using the DP@20P and AUC.

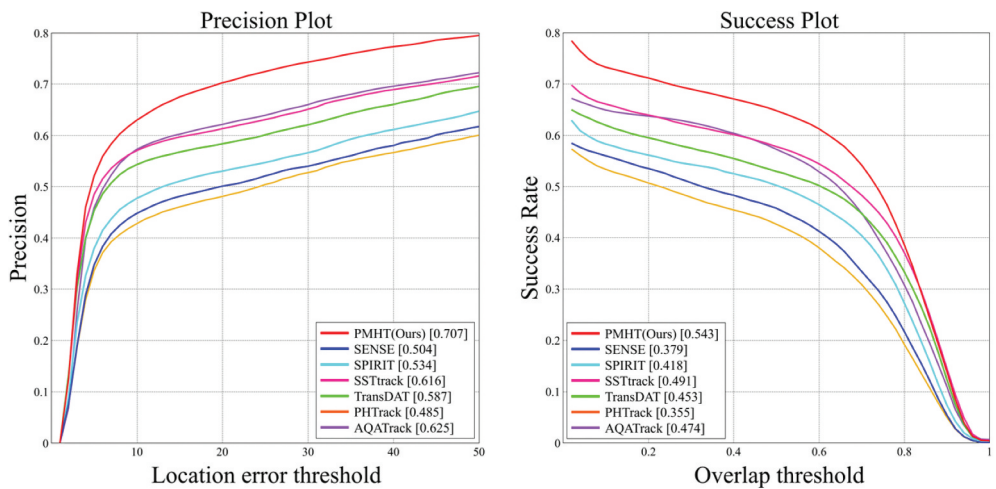


Figure 6. Quantitative performance comparison between PMHT and other SOTA HOT methods on the HOTC2024 dataset.

Table 1. Comparison of different models on HOTC2024 datasets (Bold values indicate the best results).

Model	Source	HOTC2024-ALL HOTC2024-NIR HOTC2024-RedNIR HOTC2024-VIS							
		DP@20P	AUC	DP@20P	AUC	DP@20P	AUC	DP@20P	AUC
SPIRIT 2023	TGRS2023	0.534	0.418	0.823	0.655	0.516	0.379	0.409	0.323
PHTrack 2024a	TGRS2024	0.485	0.356	0.731	0.526	0.364	0.264	0.411	0.306
AQATrack 2024	CVPR2024	0.625	0.474	0.855	0.707	0.549	0.413	0.545	0.389
TransDAT 2024	TCSVT2024	0.587	0.453	0.753	0.587	0.547	0.428	0.524	0.401
SENSE 2024b	Inform Fusion2024	0.504	0.379	0.765	0.561	0.465	0.367	0.398	0.301
SSTrack 2025b	Inform Fusion2025	0.616	0.491	0.888	0.712	0.659	0.512	0.482	0.386
PMHT(Ours)	–	0.707	0.543	0.898	0.729	0.617	0.470	0.648	0.482

Table 2. The ablation result (Bold values indicate the best results).

Baseline	PGM-ST	PGM-SP	2DML	DP@20P	AUC	FPS	FLOPs
✓				0.534	0.396	97	21.9 G
✓		✓		0.623	0.479	60	23.1 G
✓	✓			0.585	0.451	59	23.2 G
✓	✓	✓		0.651	0.500	43	24.4 G
✓	✓	✓	✓	0.707	0.543	26	26.6 G

PGM-ST: PGM’s spatio-temporal branch; PGM-SP: PGM’s spectral branch.

As shown in Table 2, the baseline model achieves a DP@20P score of 0.534 and an AUC score of 0.396, both of which are relatively low. The ablation study demonstrates that the PGM module, the 2DML module and the improvements in the loss function contribute significantly to overall performance. In particular, the final PMHT model achieves the best results, with a DP@20P score of 0.707 and an AUC score of 0.543. In terms of computational complexity and tracking speed, the introduction of the PGM and 2DML modules inevitably increases the computational overhead and leads to a decrease in inference speed. However, this is accompanied by a substantial improvement in model performance, with the DP@20P increasing by 17.3% points and the AUC by 14.7% points. Notably, the final model maintains a frame per second (FPS) of 26, which is sufficient to satisfy the real-time requirement.

5. Conclusions

In the field of hyperspectral video tracking, the proposed PFFM effectively leverages spatiotemporal context and spectral information to address the performance limitations of models under challenging scenarios such as rapid target variations and occlusions. Furthermore, by incorporating a direction-aware mechanism and a bounding box shape penalty term into the object box loss function, the model’s ability to learn the direction of predicted box offsets and enforce geometric consistency is enhanced, thereby improving both localization accuracy and generalization. Extensive validation and ablation experiments demonstrate that the PMHT model achieves superior tracking precision on the publicly available HOTC2024 dataset for hyperspectral targets.

Author contributions

CRedit: **Hongyu Ma**: Conceptualization, Data curation, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – original draft; **Linyong Xu**: Data curation, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Junping Liu**: Investigation, Software, Validation, Visualization, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

Data will be made available on request.

References

- Chen, J., K. Qian, W. Song, Y. Qiu, and S. Wang. 2025. "SiamTU: Hyperspectral Video Tracking Against Appearance Changes Using Improved SiameGAT with Adaptive Template Updating." *Infrared Physics & Technology* 147:105811. <https://doi.org/10.1016/j.infrared.2025.105811>.
- Chen, Y., Y. Tang, X. Su, J. Li, Y. Xiao, J. He, and Q. Yuan. 2024. "PHTrack: Prompting for Hyperspectral Video Tracking." *IEEE Transactions on Geoscience & Remote Sensing* 62:1–18. <https://doi.org/10.1109/TGRS.2024.3461316>.
- Chen, Y., Q. Yuan, Y. Tang, Y. Xiao, J. He, T. Han, Z. Liu, and L. Zhang. 2025. "SSTtrack: A Unified Hyperspectral Video Tracking Framework via Modeling Spectral-SpatialTemporal Conditions." *Information Fusion* 114:102658. <https://doi.org/10.1016/j.inffus.2024.102658>.
- Chen, Y., Q. Yuan, Y. Tang, Y. Xiao, J. He, and Z. Liu. 2024. "SENSE: Hyperspectral Video Object Tracker via Fusing Material and Motion Cues." *Information Fusion* 109:102395. <https://doi.org/10.1016/j.inffus.2024.102395>.
- Chen, Y., Q. Yuan, Y. Tang, Y. Xiao, J. He, and L. Zhang. 2023. "SPIRIT: Spectral Awareness Interaction Network with Dynamic Template for Hyperspectral Object Tracking." *IEEE Transactions on Geoscience & Remote Sensing* 62:1–16. <https://doi.org/10.1109/TGRS.2023.3347950>.
- Dosovitskiy, A., L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, et al. 2020. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." *arXiv Preprint arXiv: 2010.11929*. <https://doi.org/10.48550/arXiv.2010.1192>.
- Fu, H., A. Zhang, G. Sun, J. Ren, X. Jia, Z. Pan, and H. Ma. 2022. "A Novel Band Selection and Spatial Noise Reduction Method for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 60:1–13. <https://doi.org/10.1109/TGRS.2022.3189015>.
- Gan, H., X. Wang, L. He, and J. Liu. 2024. "Learned Two-Step Iterative Shrinkage Thresholding Algorithm for Deep Compressive Sensing." *IEEE Transactions on Circuits and Systems for Video Technology* 34 (5): 3943–3956. <https://doi.org/10.1109/TCSVT.2023.3325340>.
- Gao, L., P. Liu, Y. Jiang, W. Xie, J. Lei, Y. Li, and Q. Du. 2023. "CBFF-Net: A New Framework for Efficient and Accurate Hyperspectral Object Tracking." *IEEE Transactions on Geoscience & Remote Sensing* 61:1–14. <https://doi.org/10.1109/TGRS.2023.3253173>.
- Gevorgyan, Z. 2022. "Siou Loss: More Powerful Learning for Bounding Box Regression." *arXiv Preprint arXiv: 2205.12740*.
- Gu, A., and T. Dao. 2023. "Mamba: Linear-Time Sequence Modeling with Selective State Spaces." *arXiv Preprint arXiv: 2312.00752*. <https://doi.org/10.48550/arXiv.2312.00752>.
- Hou, Z., W. Li, J. Zhou, and R. Tao. 2022. "Spatial-Spectral Weighted and Regularized Tensor Sparse Correlation Filter for Object Tracking in Hyperspectral Videos." *IEEE Transactions on Geoscience & Remote Sensing* 60:1–12. <https://doi.org/10.1109/TGRS.2022.3215431>.

- Jia, M., L. Tang, B.-C. Chen, C. Cardie, S. Belongie, B. Hariharan, and S.-N. Lim. 2022. "Visual Prompt Tuning." In *European Conference on Computer Vision*, Tel Aviv, Israel, 709–727. Springer.
- Li, W., Z. Hou, J. Zhou, and R. Tao. 2023. "SiamBAG: Band Attention Grouping-Based Siamese Object Tracking Network for Hyperspectral Videos." *IEEE Transactions on Geoscience & Remote Sensing* 61:1–12. <https://doi.org/10.1109/TGRS.2023.3285802>.
- Li, Z., F. Xiong, J. Zhou, J. Lu, and Y. Qian. 2023. "Learning a Deep Ensemble Network with Band Importance for Hyperspectral Object Tracking." *IEEE Transactions on Image Processing* 32:2901–2914. <https://doi.org/10.1109/TIP.2023.3263109>.
- Liu, Z., X. Wang, Y. Zhong, M. Shu, and C. Sun. 2022. "SiamHYPER: Learning a Hyperspectral Object Tracker from an RGB-Based Tracker." *IEEE Transactions on Image Processing* 31:7116–7129. <https://doi.org/10.1109/TIP.2022.3216995>.
- Qian, K., S. Wang, S. Zhang, and J. Shen. 2025. "SiamPKHT: Hyperspectral Siamese Tracking Based on Pyramid Shuffle Attention and Knowledge Distillation." *Sensors* 23 (23): 9554. <https://doi.org/10.3390/s23239554>.
- Qian, K., J. Zhou, F. Xiong, H. Zhou, and J. Du. 2018. "Object Tracking in Hyperspectral Videos with Convolutional Features and Kernelized Correlation Filter." In *Smart Multimedia: First International Conference, ICSM 2018*, Toulon, France, August 24–26, 2018, Revised Selected Papers 1, 308–319. Springer.
- Tang, Y., Y. Liu, L. Ji, and H. Huang. 2022. "Robust Hyperspectral Object Tracking by Exploiting Background-Aware Spectral Information with Band Selection Network." *IEEE Geoscience & Remote Sensing Letters* 19:1–5. <https://doi.org/10.1109/LGRS.2022.3202039>.
- Wang, Y., X. Li, X. Yang, F. Ge, B. Wei, L. Li, and S. Yue. 2025. "Deep Feature-Based Hyperspectral Object Tracking: An Experimental Survey and Outlook." *Remote Sensing* 17 (4): 645. <https://doi.org/10.3390/rs17040645>.
- Wu, Y., L. Jiao, X. Liu, F. Liu, S. Yang, and L. Li. 2024. "Domain Adaptation-Aware Transformer for Hyperspectral Object Tracking." *IEEE Transactions on Circuits and Systems for Video Technology* 34 (9): 8041–8052. <https://doi.org/10.1109/TCSVT.2024.3385273>.
- Xiang, P., S. Qi, S. K. Jung, K. Qian, K. Cheng, S. Prasad, H. Zhou, and D. Zhao. 2026. "High-Low-Frequency Feature-Guided Diffusion Model for Hyperspectral Anomaly Detection." *Optics & Laser Technology* 203:115561. <https://doi.org/10.1016/j.optlastec.2026.115561>.
- Xie, J., B. Zhong, Z. Mo, S. Zhang, L. Shi, S. Song, and R. Ji. 2024. "Autoregressive Queries for Adaptive Tracking with Spatio-Temporal Transformers." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*: 19300–19309. <https://doi.org/10.48550/arXiv.2403.10574>.
- Xiong, F., J. Zhou, W. Charle, Y. Zhong, P. Ghamisi, and J. Chanussot. 2024. "The Hyperspectral Object Tracking Challenge (HOTC2024)." <https://www.hsitracking.com/>.
- Xiong, F., J. Zhou, and Y. Qian. 2020. "Material Based Object Tracking in Hyperspectral Videos." *IEEE Transactions on Image Processing* 29:3719–3733. <https://doi.org/10.1109/TIP.2020.2965302>.
- Yang, J., Z. Li, F. Zheng, A. Leonardis, and J. Song. 2022. "Prompting for Multi-Modal Tracking." In *Proceedings of the 30th ACM international conference on multimedia*, 3492–3500.
- Ye, B., H. Chang, B. Ma, S. Shan, and X. Chen. 2022. "Joint Feature Learning and Relation Modeling for Tracking: A One-Stream Framework." In *European Conference on Computer Vision*, Tel Aviv, Israel, 341–357. Springer.
- Zhang, J., A. T. Nguyen, X. Han, V. Q.-H. Trinh, H. Qin, D. Samaras, and M. S. Hosseini. 2025. "2dmamba: Efficient State Space Model for Image Representation with Applications on Giga-Pixel Whole Slide Image Classification." In *Proceedings of the Computer Vision and Pattern Recognition Conference*, Nashville, USA, 3583–3592.
- Zhang, Z., S. Zhang, Z. Dai, Z. Dong, and S. Zhu. 2025. "MWVOS: Mask-Free Weakly Supervised Video Object Segmentation via Promptable Foundation Model." *Pattern Recognition* 159:111100. <https://doi.org/10.1016/j.patcog.2024.111100>.

- Zhao, C., H. Liu, N. Su, and Y. Yan. 2022. "TFTN: A Transformer-Based Fusion Tracking Framework of Hyperspectral and RGB." *IEEE Transactions on Geoscience and Remote Sensing* 60:1–15. <https://doi.org/10.1109/TGRS.2022.3215816>.
- Zhu, J., S. Lai, X. Chen, D. Wang, and H. Lu. 2023. "Visual Prompt Multi-Modal Tracking." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Vancouver, BC, Canada, 9516–9526.